

Computed tomographic, magnetic resonance imaging, and ultrasonographic characteristics of retrobulbar meningiomas in 15 dogs

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Abstract

Meningioma is the most common tumor of the cranium in dogs and an important differential diagnosis for a potentially treatable disease that can be found in the peri-orbital tissues. The objective of this retrospective, case series study was to describe the CT, MRI, and US characteristics of confirmed retrobulbar meningiomas in a group of dogs. Medical records from multiple institutions were searched for canine patients with CT, MRI, and/or US imaging of a cytologically or histologically confirmed retrobulbar meningioma. Fifteen dogs met the inclusion criteria. Retrobulbar meningiomas typically appeared as a relatively well-defined conical to ovoid mass within the retrobulbar space, most often along the optic nerve and expanding the extraocular muscle cone. On CT, masses were predominantly soft tissue attenuating and variably heterogeneously contrast enhancing. While MRI features were variable, moderate to marked contrast enhancement was seen in all cases. Many of the tumors had evidence of partial mineralization, best appreciated on CT in nine patients, but also suspected based on susceptibility artifacts in three MRI cases, one of which was confirmed on CT. Regional osteolysis was a rare finding, noted in three cases, but was often accompanied by cranial cavity extension (2/3). Cranial cavity extension was also seen in the absence of regional osteolysis, identified in a total of six patients. On US, masses were echogenic and compressed the globe. The findings were consistent with previous gross and histologic descriptions and supported prioritizing retrobulbar meningioma as a differential diagnosis for dogs with the described imaging characteristics.

KEYWORDS

blind, exophthalmos, neoplasia, retrobulbar space

1 | INTRODUCTION

Meningiomas are the most common tumors of the central nervous system in dogs.^{1–3} They have been described intracranially (82%), affecting the spinal cord (15%), and less commonly, within the retrobulbar space (2–3%).^{2–4} Retrobulbar, orbital, or optic nerve sheath meningiomas arise from neoplastic transformation of the arachnoid cap cells within the optic nerve sheath or from extension of a primary

intracranial tumor.^{5,6} These masses are typically slow-growing, invading circumferentially into the surrounding connective tissues. Although most commonly benign, malignant variations have been reported.⁶ Grossly, retrobulbar meningiomas have been described as conical and adhered to the posterior aspect of the globe with a caudal taper, along the optic nerve as it approaches the optic canal.^{6,7} Characteristic histopathologic findings include large epithelioid cells arranged in tight bundles with islands of chondroid and osseous metaplasia located

around the optic nerve.⁶ The presence of osseous and cartilaginous metaplasia is otherwise very rare in canine meningiomas and has been proposed as a diagnostic aid for retrobulbar meningiomas.⁶ Treatment recommendations typically include enucleation/exenteration to remove the globe and affected tissues. Local recurrence is common and intracranial extension along the optic nerve results in infiltration of the optic chiasm and diencephalon causing blindness of the other eye.^{6–8} Radiation therapy may be recommended as it has been shown to improve survival times in canine patients with intracranial meningiomas following surgery.⁹ The median survival time for patients with retrobulbar meningiomas has not been determined but individual survival times as long as 4.5 years have been reported.^{6,7} Although rare, metastasis to the pulmonary parenchyma and heart has been reported.^{6,10,11}

Published descriptions of the imaging features of retrobulbar meningiomas are limited to textbooks and a few case reports.^{7,10,12–15} Ultrasonographically, they have been described as a poorly echogenic mass/thickening of the optic nerve, sometimes resulting in a smooth enlargement of the anterior-most portion of the optic nerve creating a smoothly curved shadow, differing from the acute-angled shadow of the normal optic nerve.^{7,12,16} Additionally, neurogenic masses may indent the posterior ocular wall with meningiomas tending to be slightly irregular in appearance.¹² CT findings include a partially mineralized, contrast-enhancing, soft tissue-attenuating mass that can be tracked to the optic nerve.¹⁵ On MRI, a well-defined mass is typically seen caudal to the globe that is hyperintense to normal cerebral gray matter on both T1 and T2 weighted images with uniform contrast enhancement.¹⁴ Signal voids may be seen within the mass, likely representing partial mineralization.¹³ On both CT and MRI, lysis/atrophy of the orbital wall and widening of the optic canal have been described.^{13,15}

Based on a review of the literature, no prior studies dedicated to the imaging features of canine retrobulbar meningiomas were available. The objective of this study was to describe the CT, MRI, and ultrasonographic characteristics of confirmed retrobulbar meningiomas in dogs (Figures 1, 2, and 3).

2 | MATERIALS AND METHODS

2.1 | Selection and description of subjects

The study was a retrospective, case series design. Electronic medical records from multiple institutions were searched for canine patients diagnosed with retrobulbar meningioma during the period of February 2007 to February 2022. Included hospitals were the Schwarzman Animal Medical Center, the Foster Hospital for Small Animals at Tufts University, Ryan Veterinary Hospital at The University of Pennsylvania, the Cornell University Companion Animal Hospital, the Small Animal Hospital at the University of Missouri Veterinary Health Center, the Veterinary Medical Center at Michigan State University,

Massachusetts Veterinary Referral Hospital and Veterinary Neurology and Imaging of the Chesapeake. An animal care and use protocol was not required and owner consent was obtained for all diagnostics. Dogs were included if a CT, MRI, or US examination was performed prior to intervention (surgical and/or radiation therapy) and if there was a histopathological or cytologic confirmation of meningioma. Final decisions for inclusion or exclusion were made by a board-certified veterinary radiologist (A.D., American College of Veterinary Radiology) and a veterinary radiology resident (A.S.).

2.2 | Data recording and analysis

Data recorded from medical records were patient signalment, presenting complaint, initial physical examination findings, methods used for pathologic confirmation of diagnosis, and treatment where available. All clinical data and studies were reviewed independently by the veterinary radiologist (A.D.) and veterinary radiology resident (A.S.). Reviewers were aware of the final diagnosis at the time of interpretation. In the event of conflicting opinions, a consensus was reached after the images were re-evaluated. The DICOM files for each case were retrieved and reviewed using a commercially based image analysis software (Asteris Keystone Software Suite) with the ability to manually level and create multiplanar reconstructions. In cases where there were multiple imaging modalities available, reviewers were allowed to see the other studies. Studies with repeat imaging following diagnosis and treatment were also evaluated for evidence of progression but not included in other descriptive analyses.

Computed tomographic studies were evaluated for location and shape of the mass, attenuation (soft-tissue or mineral), contrast enhancement (mild, moderate, or marked), contrast enhancement pattern (homogeneous or heterogeneous), presence of mineralization, appearance of the optic nerve (normal or thickened), regional osteolysis, widening of the optic canal, extension into the cranial cavity, appearance of the optic chiasm, mass effect on the brain, compression/displacement of the globe, presence of cystic components, and regional lymphadenopathy.

MRI studies were assessed for the location, shape of the mass, intensity on T2w, T2w-FLAIR, and T1w images (compared with normal cerebral gray matter), contrast enhancement (mild, moderate, or marked), contrast enhancement pattern (homogeneous or heterogeneous), evidence of restricted effusion (when DWI B1000 sequence and ADC map available), presence of susceptibility artifact, appearance of the optic nerve (thickened or normal), regional osteolysis, widening of the optic canal, extension into the cranial cavity, appearance of the optic chiasm, mass effect on the brain, compression/displacement of the globe, presence of cystic components, and regional lymphadenopathy.

US studies were evaluated for location, echogenicity of the mass compared with the surrounding musculature, evidence of blood flow on Doppler interrogation, presence of mineralization, compression of the globe, and evidence of regional osteolysis.

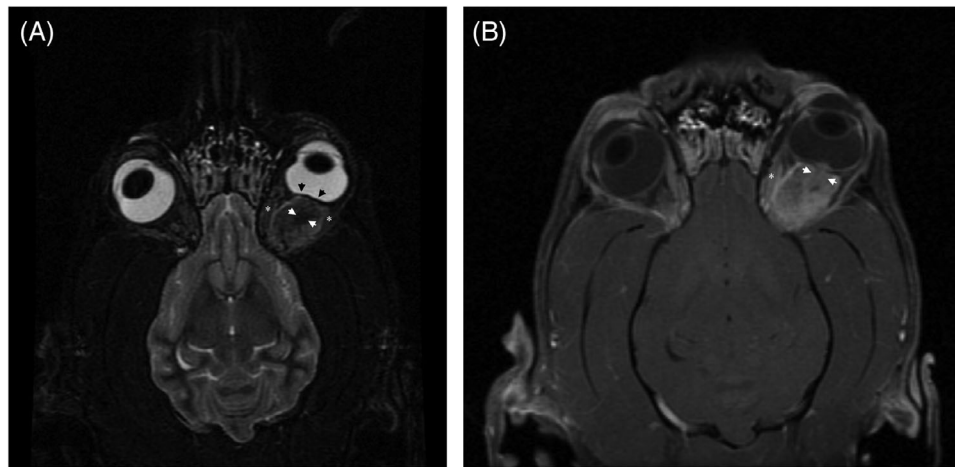


FIGURE 1 Dorsal T2w-STIR (A) and T1w postcontrast images with fat suppression (B) from case 1 demonstrate an ovoid, well-defined, left retrobulbar mass centered on the optic nerve (short white arrows) with resultant expansion and displacement of the extraocular muscle cone (white asterisks) and compression of the globe (short black arrows). Imaging parameters were not available for this study.

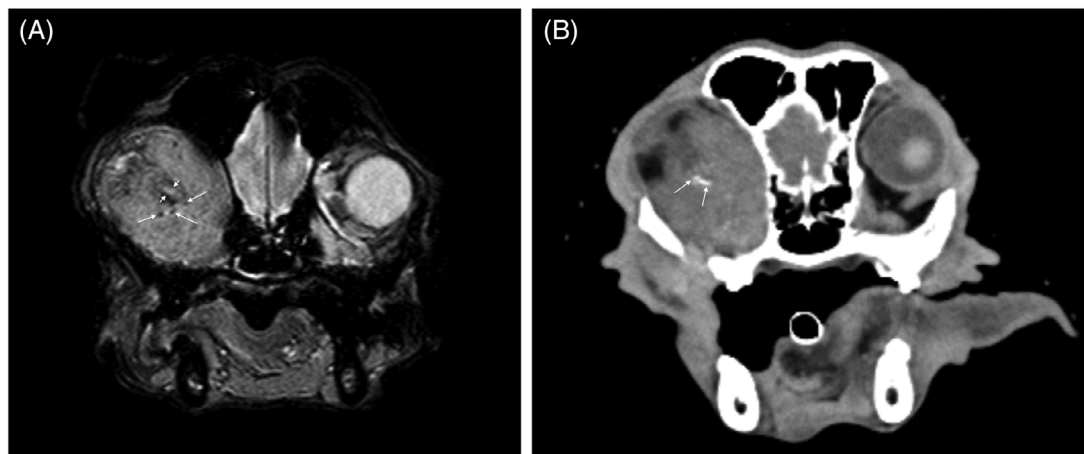


FIGURE 2 Transverse MRI and CT images from case 6, at near the same level through the well-defined, right retrobulbar mass, surrounding the optic nerve (short white arrows). On T2w-FFE images (A) there is patchy susceptibility artifact (long white arrows) within the mass, which is confirmed to represent mineralization on CT (B) demonstrated here in a soft tissue algorithm prior to contrast administration (long white arrows). Imaging parameters were not available for this study.

3 | RESULTS

3.1 | Dogs

There were 15 cases collected from 8 different institutions. Cases from different institutes had CT, MRI, and US studies using different machines and acquisition parameters, not all of which were available. The age at presentation ranged from 5 to 15 years, with a median age of 8 years. Eight dogs were female (all spayed) and seven were male (one intact and six neutered). Represented breeds include mixed ($n = 3$), Terrier mix ($n = 2$), Cocker spaniel ($n = 2$), boxer ($n = 2$), and a single Doberman pinscher, French bulldog, Pitbull, Labrador retriever, Jack Russell Terrier, and Shar Pei mix. Eleven CT examinations and seven MRI examinations were available for review. Two patients had both a CT and an MRI performed. Four cases had an ocular US performed.

3.2 | Medical record findings

Clinical findings for each of the 15 cases are detailed in Table S1. On physical examination, most patients were blind in the affected eye with an absent menace, dazzle, and pupillary light reflex or PLR ($n = 13$). The presence or absence of vision was not reported in one patient. Exophthalmos of the affected eye was also noted frequently ($n = 11$), along with decreased retropulsion ($n = 8$), third eyelid elevation ($n = 7$), and optic nerve swelling ($n = 5$). Less common findings included retinal detachment ($n = 2$), and mydriasis ($n = 2$). Scleral hemorrhage/injection, chemosis, corneal ulceration, hyperpigmentation, and neovascularization were also all noted once. Thoracic radiographs of one patient revealed the presence of a solitary pulmonary nodule, the etiology of which was not determined. The radiographs were not available for review. Enucleation or exenteration of the mass was performed in 10

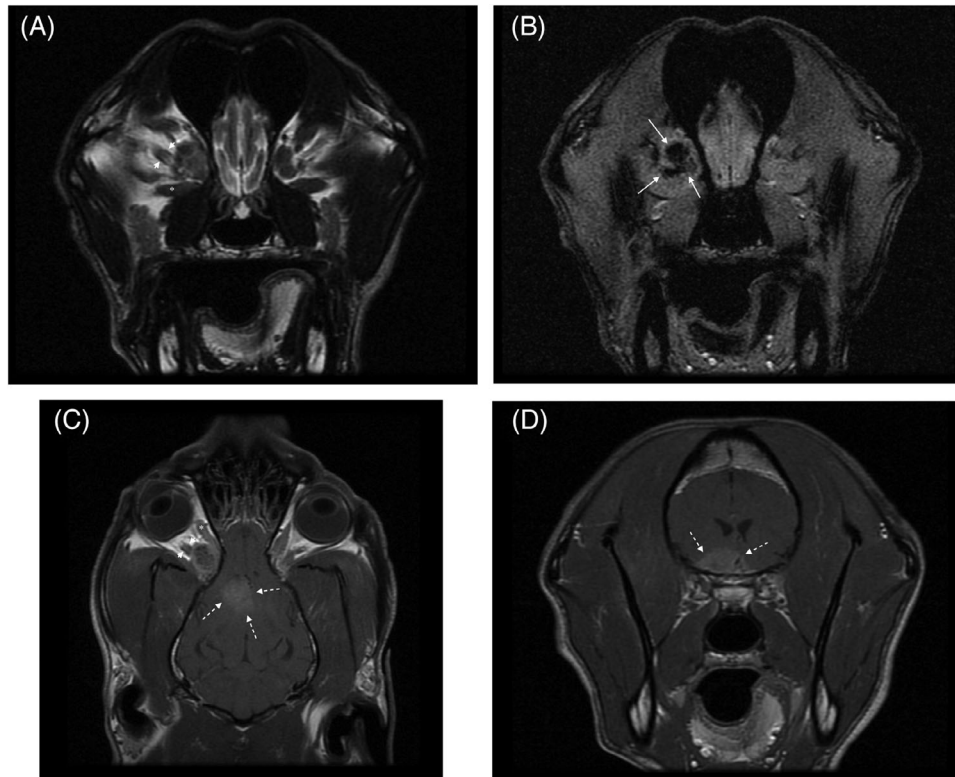


FIGURE 3 MRI images from case 13. A transverse T2w image (A) shows the right optic nerve (short white arrows) leading to the retrobulbar meningioma which causes expansion and displacement of the extraocular muscle cone (white asterisk). The transverse T2*w-GRE image (B) at the same level demonstrates marked susceptibility artifact within the retrobulbar meningioma (long white arrows). The dorsal and transverse T1w post-contrast images (C and D) show the optic nerve (short white arrows) leading through the heterogeneously contrast enhancing retrobulbar meningioma with expansion of the extraocular muscle cone (white asterisk). A uniformly contrast enhancing cranial cavity component of the mass is also seen (white dashed arrows). Imaging parameters were not available for this study.

of 15 cases with two of these cases also receiving radiation therapy. Repeat imaging was obtained in 5 of 15 cases with progressive disease seen in 3 of 5 of these cases.

3.3 | CT findings

CT findings were available for 11 of the 15 patients and are detailed in Table S2. Of these, 10 had postcontrast CT studies. The three CT studies performed at the Schwarzman Animal Medical Center were obtained using the same 64-detector third-generation CT scanner (CT Toshiba Aquilion; Canon Medical Systems) with dogs positioned in sternal recumbency and under general anesthesia. Images were acquired using a 512×512 matrix at 120 kVp and 100 mAs. Images were acquired before and after intravenous nonionic iodinated contrast material (2.2 mL/kg Omnipaque 300 mg iodine/mL, GE Healthcare) into a peripheral catheter. Images were acquired in 2.0–3.0 mm transverse images. Four patients had masses on the right and seven on the left. All masses were primarily occupying the retrobulbar space with 9 of 11 surrounding the optic nerve. The remaining 2 of 11 were lateralized but coursed along the optic nerve. All but one of the masses expanded the extraocular muscle cone. Most (10/11) masses displaced and compressed the globe resulting in variable exophthalmos. Nine of

the masses were conical in shape while three appeared ovoid; many were lobular in margination but well-defined. Seven masses were predominantly soft tissue attenuating, with 5 of 7 being homogeneous and 2 of 7 being heterogeneous. Mineralization was noted in 9 of 11 cases with four cases being predominantly mineral attenuating. Mineralization appeared to be predominantly centrally located within the mass, along the optic nerve, in 6 of 9 cases. No fluid attenuation was noted within the masses. All masses in the 10 patients administered IV contrast, displayed enhancement that varied from mild to marked, with all being heterogenous. The optic nerve could be clearly delineated in 9 of 11 cases and appeared thickened (compared with the contralateral optic nerve) in two, one of which resulted in a widened optic canal. In an additional case, the optic canal was widened, however, the optic nerves appeared bilaterally symmetric/normal. Regional osteolysis, predominantly along the medial wall of the orbit, was noted in 3 of 11 cases, two of which demonstrated extension into the cranial cavity through lytic regions. Two additional cases without regional osteolysis had extension of the mass into the cranial cavity. In these cases, extension into the cranial cavity was suspected along the optic nerve as it was thickened in both, resulting in widening of the optic canal in one. All cases with extension into the calvarium had a variable secondary mass effect on the adjacent brain parenchyma. The portion of the mass within the calvarium was predominantly isoattenuating to the brain parenchyma

with strong, uniform contrast enhancement. Three of the four cases with intracranial extension had an abnormal appearing optic chiasm, one appearing mildly thickened and the other two being more mass-like with subsequent mass effect on the brain. Six of the eleven cases had mild regional lymphadenopathy, affecting the ipsilateral medial retropharyngeal (3/6), superficial cervical (1/6), or mandibular (1/6) lymph node. Another case of mild regional lymphadenopathy affected the contralateral medial retropharyngeal lymph node. (See Tables 1, 2 and 3).

3.4 | MRI findings

Seven patients had MRI examinations performed; however, one study was excluded as the MRI was performed following surgical intervention. The MRI findings for each of the remaining six cases are detailed in Table S3. Two of these cases also had a CT examination performed. The sole MR study of the brain performed at the Schwarzman Animal Medical Center was acquired with the patient under general anesthesia using a 1.5 Tesla magnet and a human knee coil (Achieva; Philips Medical Systems), before and after intravenous injection of gadolinium-based contrast medium, gadodiamide (Omniscan; GE Healthcare). The study had a slice thickness of 2.5, 2.75, or 3.0 mm. T2-weighted sagittal images were acquired with a time to repetition (TR) of 4193 ms and time to echo (TE) of 85 ms. T2 weighted transverse images had a TR of 3500 ms and a TE of 85 ms; Proton density-weighted sequences had a TR of 2500 ms and a TE of 15 ms, T2-FLAIR weighted sequences had a TR of 6000 ms and a TE of 120 ms and T2*FFE sequences had a TR of 900 ms and TE of 9 ms. T1-weighted transverse images before and after intravenous contrast administration had a TR of 372 ms and TE of 20 ms. The dorsal postcontrast T1-weighted sequences had a TR of 572 ms and a TE of 11 ms. The major MRI findings are outlined in Table 4. Of the six cases for which MRI studies were reviewed, four had a mass on the left and two had a mass on the right. Exophthalmos and subsequent compression of the globe was noted in all cases (6/6). Masses were ovoid (3/6) to conical (3/6) with one of the conical masses being additionally lobular. All masses were well-defined and surrounded the optic nerve, expanding the extraocular muscle cone. On the T2w images, masses were hyperintense (2/6), hypointense (2/6), or isointense (2/6) when compared with normal cerebral gray matter. Most masses were heterogeneous/had mixed intensity (4/6). When T2w-FLAIR sequences were available (4/6 cases), masses were isointense (2/4), hyperintense (1/4), or hypointense (1/4). Most were heterogeneous in intensity (3/4). On T1w sequences (available for 5/6 cases), masses were isointense (3/5), hypointense (1/5), or minimally hyperintense (1/5). Again, most were heterogeneous in T1w intensity (4/5). All masses in patients administered IV contrast (5/6) were moderately to markedly contrast enhancing with 4 of 5 cases being heterogeneous and 1 of 5 being homogeneous. A T2*w GRE sequence was available in 4 of 6 cases, which demonstrated susceptibility artifact in 3 of 4 cases, one of which had mineralization within the mass confirmed on CT. In two of these cases, susceptibility artifact appeared to be predominantly along the optic nerve. Diffusion-weighted imaging

was available for 4 of 6 cases and demonstrated regions of restricted diffusion within the mass in all cases although the masses were difficult to delineate. The optic nerve was visualized in all (6/6) and appeared variably thickened (when compared with the contralateral optic nerve) in 3 of 6 cases. Three cases demonstrated extension of the mass into the ventral calvarium resulting in a variably thickened and lobular optic chiasm. Cranial cavity extension was suspected to be along a thickened optic nerve in two of these cases, one of which had slight enlargement of the optic canal. In the last case, extension through regions of cranial lysis, along the medial wall of the orbit, was suspected. Regional meningeal contrast enhancement was noted in that case. The portion of the masses within the calvarium was variable in size but broad-based in all cases with a dural tail sign appreciated in 2 of 3. On T2w images, the intracranial portion of the mass was isointense (2/3) or hyperintense (1/3) to normal gray matter. A T2w-FLAIR sequence was available in two cases in which the mass was isointense in both, one of which had an ill-defined rim of peripheral hyperintensity, likely representing perilesional edema. Masses were isointense to slightly hyperintense on T1w images with strong uniform contrast enhancement. Mild lymphadenopathy was identified in 2 of 6 cases with one case having ipsilateral mandibular lymphadenopathy and the other having contralateral medial retropharyngeal lymphadenopathy.

3.5 | US findings

Ultrasound examinations were available in four patients, one with disease on the right and the remaining three on the left. The US findings for each case are detailed in Table S4. Evaluation of the masses by US was often limited. Two US studies of the eye were acquired using the same US system at the Schwarzman Animal Medical Center (Aplo i700, Canon Medical Systems, Tustin, CA) using a 17 MHz Hockey Stick linear transducer and a Multi-frequency Ultra-wideband iDMS Linear transducer with a frequency range of 4.0–18.0 MHz. The main findings are in Table 3. An echogenic mass was identified in all cases and in 2 of 4 cases, appeared to surround the optic nerve. Three masses were hypoechoic when compared with the extraocular musculature, while one was isoechoic. Two of the four masses were heterogeneous and two were relatively homogeneous. Doppler interrogation was used to confirm blood flow within 3 of 4 masses. Mineralization was not detected in any of the masses, despite two having confirmed partial mineralization on CT. All masses resulted in compression of the globe. No evidence of regional osteolysis was identified.

4 | DISCUSSION

This is the first published report focused on the imaging features of retrobulbar meningiomas/optic nerve sheath meningiomas in dogs. Masses most commonly appeared as a relatively well-defined, often lobular, conical to ovoid mass conforming to the retrobulbar space. All except two masses appeared to surround the optic nerve. The other two masses were adjacent to but coursing with the optic nerve. In all

TABLE 1 Patient information.

	Signalment	Presenting complaint	Physical/ophthalmic examination	Blind?	Treatment/Outcome
Case 1	8Y FS Cocker Spaniel	Retrobulbar mass with vision loss	OS: Exophthalmos with decreased retropulsion	Yes, OS	Surgery
Case 2	12.5Y MC Mixed breed	Subconjunctival hemorrhage	OD: Exophthalmos, scleral hemorrhage, hyperemia and fibrosis, epiphora, hypermature cataract, protrusion and elevated nictitans; Left head tilt (historic)	Yes, OD	Surgery. Repeat CT examination 1.5 months later revealed progressive disease. Euthanized later on life thought to be unrelated to the retrobulbar mass
Case 3	8.5Y FS Terrier mix	Exophthalmos and elevated nictitans	OD: Exophthalmos, hyperemia, elevated nictitans, retinal detachment, optic nerve hyperemia, absent PLR, dazzle and menace	Yes, OD	Treated with radiation therapy. Repeat CT 7.5 months later revealed progressive disease
Case 4	15Y FS Terrier mix	Exophthalmos	OS: Exophthalmos, decreased retropulsion, absent menace and PLR	Yes, OS	Exenteration. Repeat CT 2 months later revealed stable disease
Case 5	7Y FS Boxer	Corneal ulcer referral	OS: Exophthalmos, hyperemia, chemosis decreased retropulsion, absent menace and dazzle, elevated nictitans, and corneal ulcer	Yes, OS	None
Case 6	5Y MC French Bulldog	Exophthalmos and third eyelid prolapse	OD: Exophthalmos	N/A	None
Case 7	8.5Y FS Pitbull terrier	Sudden onset unilateral blindness	OS: Exophthalmos, dorsolateral deviation, decreased retropulsion, hyperemic, absent menace, decreased PLR, elevated third eyelid, and mild mydriasis	Yes, OS	None
Case 8	8Y FS Mixed breed	Third eyelid elevation	OS: Exophthalmos, absent menace, elevated third eyelid raised, ulceration of the third eyelid surface	Yes, OS	Exenteration with partial zygomectomy. Repeat CT 8 months later revealed a destructive ipsilateral rhinopathy
Case 9	5Y MI Lab Retriever	Abnormal vision	OS: Enophthalmos, absent menace and PLR, elevated third eyelid, atrophy of the optic nerve head OS	Yes, OS	Surgical exenteration
Case 10	7Y FS Boxer	Suspect retrobulbar mass	OS: Exophthalmos, decreased retropulsion, anterior displacement of the posterior aspect of the globe and optic nerve, absent menace and PLR	Yes, OS	Surgical exenteration
Case 11	13Y MC Jack Russell Terrier	Buphthalmia and exophthalmos	OS: Buphthalmia, decreased retropulsion, absent menace, dazzle and PLR, scleral injection, marked corneal hyperemia, corneal hyperpigmentation and neovascularization. OD: lower eyelid mass (1 mm)	Yes, OS	Surgical exenteration
Case 12	6Y MC Mixed breed	Optic nerve tumor	OD: Exophthalmos, decreased retropulsion, absent PLR, dazzle and menace, retinal detachment and edematous optic nerve head OD	Yes, OD	Enucleation followed by radiation therapy. Repeat CT 3 months later revealed progressive disease

(Continues)

TABLE 1 (Continued)

	Signalment	Presenting complaint	Physical/ophthalmic examination	Blind?	Treatment/Outcome
Case 13	11Y MC Shar Pei mix	Progressive vision loss	OU: Absent dazzle, menace and PLR OD: Severe optic nerve swelling	Yes, OU	Enucleation
Case 14	10Y MC Cocker Spaniel	Referral for an orbital mass with optic nerve involvement	OS: Exophthalmos, decreased retropulsion, hyperemia, absent menace, PLR and dazzle, mydriasis, severe optic nerve swelling	Yes, OS	Enucleation
Case 15	5Y FS Doberman	Horner's syndrome with elevated 3rd eyelid	OS: Enophthalmos, painful on palpation, intact PLR and dazzle, optic nerve swelling	No	Enucleation

Abbreviation: PLR, pupillary light reflex.

but one case, the masses resulted in the expansion of the surrounding extraocular muscle cone. Mineralization of the mass was also a common finding on both CT and suspected on MRI.

Prior literature that has detailed the appearance of intracranial meningiomas on CT and MRI is available.^{1-3,17-19} On CT, masses tend to be isoattenuating to hyperattenuating to the brain parenchyma with strong contrast enhancement.^{2,17,19} On MRI, masses tend to be hyperintense on T2w/T2w-FLAIR images and hypointense to isointense on T1w images, when compared with normal cortical gray matter, with strong contrast enhancement.^{1-3,18,19} Retrobulbar meningiomas had many imaging characteristics comparable to intracranial meningiomas. On CT, they were typically predominantly soft tissue attenuating, isoattenuating to the brain parenchyma, and heterogeneously contrast enhancing. While MR features were variable, moderate to marked contrast enhancement was seen in all cases. In one case the mass was slightly T1 hyperintense (compared with normal gray matter), a definitive cause for which was not identified. In general, hyperintensity on T1W images could be secondary to calcification, melanin, protein-containing lesions, vasopressin, fat, and potentially myelin degradation products.²⁰

Mineralization is not common within intracranial meningiomas but has been reported.^{1-3,17-19} This, however, is commonly reported in retrobulbar meningiomas and likely relates to the presence of islands of osseous metaplasia seen histologically.^{6,13,15} Many of the tumors in these cases had evidence of partial mineralization, which was best appreciated on CT in 9/11 patients but was also suspected based on susceptibility artifact in 3 of 6 MRI studies (one of which was confirmed on CT). Mineralization varied in severity between cases but in most appeared central, along the course of the optic nerve. While the presence of mineralization within a retrobulbar mass remains nonspecific, previous reviews of retrobulbar disease only reported it in neoplastic conditions, consistent with previous literature.²¹ Comparison of the severity or pattern of mineralization in retrobulbar meningiomas, compared with other neoplastic conditions, could be useful for further differentiation between different neoplasms.

Regional osteolysis was a rare finding, noted in 3 of 15 cases. This finding has previously been found more often with neoplasia rather than infectious/inflammatory disease.²¹ Cranial cavity extension was

identified in a total of 6 patients. While extension was suspected across regional osteolysis in two cases, it appeared to be along the optic nerve in the remaining four. The appearance of the intracranial portion of the mass was comparable to prior descriptions of intracranial meningiomas.^{1-3,17-19}

On US, three of the four masses were predominantly hypoechoic to the surrounding extraocular musculature, and one was isoechoic. All masses resulted in the displacement of the globe. These findings are consistent with what has been previously described regarding ultrasonographic findings with retrobulbar masses.^{12,22-24}

The age of the affected animals in this series was comparable to a prior large retrospective study⁶ and a breed or sex predilection was not identified. The clinical signs noted in these cases were also consistent with what is reported in the literature, with exophthalmos, decreased retropulsion, elevation of the third eyelid, and blindness being the most common.^{6,25} Almost all patients (13/15) were blind in the affected eye; however, vision was not reported in one patient.

Additional differentials for a retrobulbar mass include abscessation, granuloma formation, and other neoplasms including carcinoma and sarcoma, with carcinoma most commonly extending from the nasal cavity.^{21,26,27} Findings in this study can aid in differentiation from other retrobulbar diseases. Fluid attenuation, a common finding in retrobulbar abscesses, was not seen in any of the cases in this series. Mineralization, most often present with neoplasia, was variable but in some cases was marked, as a subjectively unique feature, which may aid in prioritizing meningioma. While osteolysis can commonly be seen with other neoplasms, it was a rare finding in this series, even in spite of cranial cavity extension in some cases.^{21,23,24} Finally, the expansion of the extraocular muscle cone was a notable finding that had not been previously described in this way. This in combination with abnormalities of the optic nerve and optic chiasm further aid in the association of these masses with the optic nerve, aiding differentiation from other retrobulbar neoplasia.

There were several limitations of this study, many of which were related to the retrospective nature. The image quality across several different modalities was quite variable and sequences between different institutes were not standardized. The small sample size of this study is another limitation. Furthermore, the multicenter nature of

TABLE 2 CT Findings.

Laterality	Location	Expansion of the extraocular musculature		Shape	Predominant attenuation	Heterogeneous or homogeneous	Mineralized	Contrast enhancement pattern	Optic nerve visualization	Widening of the optic canal			Intra-calvarial extension	Regional lymphadenopathy
		Yes	No							Optic nerve appearance	Yes	No		
Case 2	OD	Surrounding optic nerve	Yes	Conical, lobular	Soft-tissue, isoattenuating to brain	Homogeneous	No	Mild, heterogeneous	No	N/A	No	Yes	Yes	Right MRP
Case 3	OD	Surrounding optic nerve	Yes	Conical, lobular	Soft-tissue, isoattenuating to brain	Homogeneous	yes	Marked, heterogeneous	Yes	Thickened	No	No	Yes	Right SC
Case 4	OS	Along optic nerve, lateralized	Yes	Conical	Soft-tissue, isoattenuating to brain	Homogeneous	Yes	Mild, heterogeneous	Yes	Normal	No	No	No	No
Case 5	OS	Surrounding optic nerve	Yes	Conical, lobular	Soft tissue isoattenuating to brain	Heterogeneous	Yes	Moderate, heterogeneous	No	N/A	No	No	No	No
Case 6	OD	Surrounding optic nerve	Yes	Conical	Soft-tissue, isoattenuating to brain	Heterogeneous	Yes	Marked, heterogeneous	Yes	Normal	No	Yes	Yes	Right MRP
Case 7	OS	Surrounding optic nerve	Yes	Ovoid	Mineral	Heterogeneous	Yes	Mild, heterogeneous	Yes	Normal	Yes	No	No	No
Case 8	OS	Along optic nerve, lateralized	No	Conical	Mineral	Heterogeneous	Yes	Marked, heterogeneous	Yes	Normal	No	Yes	No	Left MRP
Case 9	OS	Surrounding optic nerve	Yes	Ovoid	Mineral	Homogeneous	Yes	N/A	Yes	Normal	No	No	No	No
Case 10	OS	Surrounding optic nerve	Yes	Ovoid, lobular	Mineral	Heterogeneous	Yes	Marked, heterogeneous	Yes	Normal	No	No	No	No
Case 11	OS	Surrounding optic nerve	Yes	Conical	Soft-tissue, isoattenuating to brain	Homogeneous	No	Marked, heterogeneous	Yes	Normal	No	No	No	Left MAN
Case 12	OD	Surrounding optic nerve	Yes	Conical, lobular	Soft-tissue, isoattenuating to brain	Homogeneous	Yes	Moderate, heterogeneous	Yes	Thickened	Yes	No	Yes	Left MRP

Abbreviations: MAN, mandibular lymph node; MR, medial retropharyngeal lymph node; SC, superficial cervical lymph node.

TABLE 3 MRI Findings.

Later-ality	Location	Expansion of extraocular musculature	Shape	T2 intensity	T2-FLAIR intensity	T1 intensity	Contrast enhancement pattern	Restricted diffusion	GRE/FFE susceptibility artifacts	Optic nerve visualized	Optic nerve appearance	Widening of the optic canal	Regional osteolysis	Intracanal extension	Regional lymphadenopathy
Case 1	OS	Surrounds optic nerve	Conical, lobular	Heterogeneously hyperintense	Heterogeneously isointense	Heterogeneously hyperintense	Marked heterogeneous	N/A	N/A	Yes	Thickened	Yes	No	No	No
Case 6	OD	Surrounds optic nerve	Conical	Heterogeneously isointense (RB); Homogeneously hyperintense (IC)	N/A	Heterogeneously isointense (RB); Homogeneously isointense (IC)	Marked heterogeneous	N/A	Minimal	Yes	Normal	No	Yes	Yes	No
Case 9	OS	Surrounds optic nerve	Ovoid	Homogeneously hypointense	N/A	N/A	N/A	Yes	N/A	Yes (partial)	Normal	No	No	No	No
Case 13	OD	Surrounds optic nerve	Ovoid	Heterogeneously hypointense (RB); Homogeneously isointense (IC)	Heterogeneously hypointense (RB); Homogeneously isointense with peripheral hyperintensity (IC)	Heterogeneously hypointense (RB); Heterogeneously isointense (IC)	Marked heterogeneous	Yes	Marked	Yes	Variably thickened	No	No	Yes	No
Case 14	OS	Surrounds optic nerve	Conical	Heterogeneously hyperintense; isointense	Heterogeneously hyperintense	Heterogeneously isointense	Marked heterogeneous	Yes	Mild	Yes (partial)	Thickened	Yes	No	Yes	Left MAN
Case 15	OS	Surrounds optic nerve	Ovoid	Homogeneously isointense (RB); Homogeneously isointense (IC)	Homogeneously isointense (RB); Homogeneously isointense (IC)	Homogeneously isointense (RB); Homogeneously isointense (IC)	Moderate homogeneous	Yes	No	Yes	Normal	No	No	No	Right MRP

Abbreviations: CC, Cranial cavity; MAN, mandibular lymph node; MRP, medial retropharyngeal lymph node, RB, retrobulbar.

TABLE 4 Ultrasound findings.

	Eye affected	Echogenicity to surrounding soft-tissue	Heterogeneous or homogeneous	Flow on Doppler	Mineralized	Surrounding the optic nerve	Regional bony lysis
Case 2	OD	Hypoechoic	Heterogeneous	Yes	No	Yes	No
Case 4	OS	Hypoechoic	Heterogeneous	Yes	No	Unsure	No
Case 5	OS	Isoechoic	Homogeneous	No	No	Unsure	No
Case 11	OS	Hypoechoic	Homogeneous	Yes	No	Yes	No

this study resulted in variability in the availability of medical records. Additional limitations include abnormalities noted on physical examination or during the imaging study that were not further followed up on cytologically or histopathologically, such as a mild lymphadenopathy or pulmonary nodule in one case. Additional studies further investigating the imaging features highlighted by this study and comparing them with other retrobulbar neoplasms could aid in further differentiation.

In conclusion, the current study found that common imaging characteristics of retrobulbar meningiomas included partial mineralization, conical to ovoid shape, location surrounding the optic nerve, expansion of the extraocular muscle cone, and cranial cavity extension either along the optic nerve or through regional lysis. Retrobulbar meningiomas should therefore be prioritized as a differential diagnosis for dogs with these imaging findings.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- Conception and design: Sidhu, DeJesus
- Acquisition of data: Sidhu, DeJesus
- Analysis and interpretation of data: Sidhu, DeJesus

Category 2

- Drafting the article: Sidhu, DeJesus
- Revising the article for intellectual content: Sidhu, DeJesus

Category 3

- Final Approval of the Completed Article: Sidhu, DeJesus

Category 4

- Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: Sidhu, DeJesus

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available as supplementary files and from the corresponding author upon request.

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The authors followed PROCESS reporting guidelines.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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